P-R-201b
MARCH 29, 1961
SUPERSEDING
Int. Fed. Spec. P-R-00201a(GSA-PBS)
April 28, 1960

FEDERAL SPECIFICATION

REMOVER, WATER EMULSION TYPE FLOOR WAX

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers commercial wax remover for use in stripping commercial water emulsion floor wax from asphalt tile, rubber tile, and linoleum floor surfaces. Type III wax remover is especially designed for hospital use.

1.2 Classification.—The compound covered by this specification shall be of one grade and of the following types, as specified:

Type I.—Regular concentrate. (Mix 1 part of concentrate with 8 parts of water.)

Type II.—High concentrate. (Mix 1 part of concentrate with 32 parts of water.)

Type III.—High concentrate without volatile ammonia. (Mix 1 part of concentrate with 32 parts of water.)

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and Standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

L-S-626—Sponges; Cellulose Type.

SS-T-306—Tile, Floor, Asphalt.

ZZ-T-301—Tile; Floor, Rubber.

LLL-L-367 — Linoleum; Plain, Jaspe and Marbleized.

PPP-C-96—Cans, Metal, 28 Gage and Lighter.

PPP-D-729—Drums; Metal, 55-Gallon (For Shipment of Noncorrosive Materials).

PPP-P-704—Pails: Shipping, Steel (1 Through 12 Gallon).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.—The compound shall be a homogeneous, free flowing liquid. It shall not contain any odor that may be classified as objectionable or cause skin irritations when used in accordance with the manufacturer's directions (see 4.4.3).

3.1.1 Type III material shall contain no volatile ammonia (see 4.4.7).

3.2 Stability in storage.—The compound shall be stable and not lose its original effectiveness or otherwise deteriorate when stored for 9 months in a closed shipping container at room temperature (see 6.4). Material susceptible to decomposition will be required to contain a preservative or inhibitor to comply with the specified storage life (see 5.1.1.3).

3.3 Miscibility.—The compound shall be completely miscible with distilled water when tested as specified in 4.4.2.

3.4 Rinsing properties. — The compound shall be free rinsing when tested as specified in 4.4.4.

3.5 Wax removal efficiency. — The floor wax removal compound shall remove 95 percent of an aged dyed wax film from a clean finished linoleum surface as determined by instrumental measurement. The reflectances of light-colored standard linoleum panels, of the aged dyed wax surface, and of the surface after treatment with wax removal compound, shall be measured in a Hunter Multi-Purpose Reflectometer, as described in 4.4.5.

3.6 Deleterious action on linoleum, asphalt tile and rubber tile floors.—The surfaces of linoleum, asphalt tile and rubber tile shall undergo no deleterious attack by the wax removal compound when tested as described in 4.4.6. Objectionable deleterious actions include softening, blistering, swelling, cracking, pitting, solvent action, and any appreciable color changes by staining, film formation, darkening, bleaching, oxidation, or any visual changes in these floor surface panels, as described in 4.4.6.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection responsibility. — The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Inspection and tests of components.*— In accordance with 4.1, it is the supplier's responsibility to insure that all components and materials have been tested to meet the requirements of this as well as subsidiary specifications referenced herein.

4.2 Inspection of the filled containers.—An examination will be made of filled containers

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in accordance with the classification of defects listed below. The lot size for the purpose of determining the sample size shall consist of all containers of the same type and capacity filled with material from the same manufactured batch and offered for delivery at one time. The sample unit for examination shall be one filled container. The sample size shall be level L–8 of Military Standard MIL–STD–105. The acceptable quality level (AQL) shall be 4.0 expressed as defects per 100 units.

Examine	Defect
Markings	Omitted, incomplete, incorrect, illegible, not in accordance with requirements (see 5.3.3).
Container	Not type or size specified. Not exterior coated (as applicable). Not olive drab exterior coat (as applicable). Leaks. Badly dented. Crushed. Not properly lined (as applicable).
Closure	Not in accordance with requirements. Not prescribed nozzle, spout or spout closure (as applicable). Spout, nozzle or closure leaks.
Fill	Less than specified. Overfilled with insufficient space for thermal expansion.

4.2.1 *Examination of preparation for delivery.*—An examination will be made to determine that markings, packaging, packing, contents and weights comply with the requirements of section 5 of this specification. Defects will be scored in accordance with the classification below. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the inspection lot. The inspection level shall be L–4 of Military Standard MIL–STD–105 and the acceptable quality level (AQL) shall be 4.0 expressed as defects per 100 units.

Examinations	Defects
Markings (as applicable)	Incorrect; illegible; omitted; of improper size, location, sequence, or method of application.
Materials (as applicable)	Any nonconforming component; component missing, damaged, or otherwise defective, affecting serviceability.
Workmanship	Inadequate application of components such as: Loose or inadequate strapping; bulged or distorted containers.

4.3 Tests of the end item.—The end item shall be tested for each of the characteristics in table I. For these tests a composite sample shall be used to represent the inspection lot. The composite sample shall consist of a one pint specimen each taken from the randomly selected filled containers, at inspection level L–3 determined from the lot size in accordance with Military Standard MIL–STD–105. The number of containers to be sampled shall be not less than four. The composite sample specimens shall be placed in a clean, dry glass or plastic container, thoroughly mixed, properly labeled for use in tests of the end item. All requirements are applicable to the composite sample unit and no failures shall be permitted.

4.4 Test procedures.

4.4.1 The composite sample prepared as specified in 4.3 shall be tested for each of the characteristics specified in table I. The number of determinations shall be as specified therein.

4.4.2 *Miscibility.*

4.4.2.1 *Type I.* — To 400 ml. of distilled water in a 500-ml. graduate add 50 ml. of the compound. The compound shall be a free flowing liquid. Cover the graduate, invert four times and immediately examine the column of liquid in the graduate against a strong light for any lack of homogeneity.

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TABLE I.—*Tests of the end item*

Characteristic	Specification requirement	Reference test method	Number of determinations per composite sample	Results reported as pass or fail[1]
Miscibility	3.3	4.4.2	2	X
Objectionable odor	3.1	4.4.3	2	X
Rinsing properties	3.4	4.4.4	2	X
Wax removal efficiency	3.5	4.4.5	1	X
Deleterious action	3.6	4.4.6	1	X
Volatile ammonia type III only	3.1.1	4.4.7	1	X
Storage stability[2]	3.2	6.4	--	

[1] If failure is indicated report either description of failure or numerical point of failure, as applicable.
[2] The conditions set forth in 6.4 shall apply.

4.4.2.2 *Types II and III.*—To 320 ml. of distilled water in a 500-ml. graduate add 10 ml. of the compound. The compound shall be a free flowing liquid. Cover the graduate, invert four times and immediately examine the column of liquid in the graduate against a strong light for any lack of homogeneity.

4.4.3 *Objectionable odor.*—There shall be no objectionable odor at a distance of 3 feet from an open container holding a solution of the compound as prepared for the miscibility test. Except for type III, a faint odor of ammonia shall not be considered objectionable.

4.4.4 *Rinsing properties.*

4.4.4.1 *Synthetic hard water.* — The synthetic hard water shall be made to the following formula:

0.264 grams calcium chloride, $CaCl_2 \cdot 2H_2O$

0.295 grams magnesium sulfate, $MgSO_4 \cdot 7H_2O$

The hard water salts should be dissolved in a small portion of distilled water, the solution transferred to a liter volumetric flask, and the flask filled to graduation.

4.4.4.2 *Procedure.*—Dissolve 10 ml. of type I in 80 ml. of synthetic hard water or 3 ml. of types II and III in 96 ml. of synthetic hard water 25° ± 1°C. in a clean 250-ml. Erlenmeyer flask. Stopper and shake vigorously for 1 minute. Pour out the solution. Rinse the flask by the same procedure, using three separate 100-ml. portions of synthetic hard water alone. Invert the flask, allow to dry, and examine for any residue not rinsed from the interior. The flask shall contain no more residue than a similar flask allowed to dry after rinsing with hard water alone.

4.4.5 *Wax removal efficiency.*

4.4.5.1 *Washability apparatus.*[1]—The apparatus consists essentially of an electric motor mounted on a flat metal plate and a mechanism through which the motor imparts a reciprocating motion to a sponge, held in a metal box, lengthwise across a test panel, clamped to a tray located on a platform situated above the mechanism.

4.4.5.1.1 The power is transmitted from the motor to a driving arm by means of a belt,

[1] This can be purchased from the Gardner Laboratory, Inc., P.O. Box 5728, Bethesda 14, Md.

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pulley, and gear reduction box. A flexible wire, guided by a pulley at each end of the platform, runs from the driving arm to each end of the sponge box. Tension on the wire is just sufficient to take up any slack. The path of travel over the test surface is over the width of the test panel. The sponge box is 3¾ inches long, 3 inches wide, and 1 inch deep.

4.4.5.1.2 The sponge, which shall conform to type I of Federal Specification L–S–626, is preshrunk and cut to a dry size of about 3¾ by 2⅞ by 1½ inches. The total weight of the sponge box and dry sponge shall be 1 pound plus or minus ½ ounce. Sheet lead may be fastened to the top of the sponge box to attain this weight, if necessary.

4.4.5.2 *Preparation of standard test panels.*

4.4.5.2.1 New white linoleum panels, Armstrong pattern No. 23—plain white, heavy duty, battleship grade—6½ inches x 6½ inches shall be mounted on 6 inches x 6 inches x ⅛ inch extra duty glass panels with a colorless water-resistant adhesive (General Electric Glyptal No. 1276 and Dupont Duco Cement are suitable). It is recommended that the edges of the glass panels be polished to prevent the hands from being cut in the handling of wet panels. The porous linoleum backing surface and one side of the glass are given a coat of adhesive. Glyptal No. 1276 is applied easier by brush, when it is thinned with Glyptal 1511–M Thinner. The linoleum and glass pieces are pressed together. A stack is formed with the linoleum part of the panel facing down, and a weight is kept on the stack overnight. Two coats of adhesive may be used when this is considered necessary. Solvent in the adhesive should not be allowed to come in contact with the top linoleum surface. Panels, which have a changed test surface by direct contact with the adhesive or by the bleed through effect, shall immediately be rejected for use in these tests. The half-inch over-lap of the linoleum serves to catch the excess adhesive, which is squeezed out by the weight, and also gives a much neater panel when the edges are trimmed. The following day the edges are trimmed by running a curved linoleum knife along the side of the glass with a slight angle away from the glass side in order to keep any pressure off the edge of the glass, which may cause chipping. Stand the trimmed panels edgewise, with an air space between them in an oven at 50° ± 2°C. for 24 to 48 hours. Satisfactory panel racks can be prepared with two rows of anchored ceiling nails three to five inches apart on a wood plank with a quarter inch interval between the nails in each row. This step serves three useful purposes. The adhesive becomes more resistant to subsequent soaking procedures, and the top surface finish becomes more uniform and more resistant to removal solutions so that better overall results are obtained. Some volatiles which crystallize the surface wax coat when they bleed through the linoleum are expelled from the adhesive at this elevated temperature. Bleed-through is more predominant on standard linoleum in stacked positions. Bleed-through is not often encountered on battleship linoleum.

After they have cooled to room temperature, the mounted panels are soaked in a 5 percent potassium oleate soap solution containing 2 ml. of 28 percent NH_3 per liter for five minutes. The panels shall be scrub sponged and rinsed repeatedly until running water readily drains from all parts of the linoleum surface in small droplets. When satisfactory drainage is attained, the panel is blotted with the squeezed out sponge and allowed to dry in a vertical drain rack at room temperature. This liberal drainage indicates that the surface of the linoleum is satisfactory for the following procedures. However, if a puddling effect occurs upon application of the dyed wax, the panel should be rejected and cleaned for reuse. Linoleum panels may be reused several times when they are properly processed and the results will be comparable with new panels. Considerable time is consumed in panel preparation and a great advantage is realized when the panels can be reused. Reuse of a panel is prohibited: (a)

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when the surface is badly scratched, chipped or marred; (b) when all of the wax is not removed by two cleansing operations which follow the efficiency test, and (c) when loosening of the linoleum from the plate glass is apparent. Linoleum panels for reuse are soaked for 5 minutes in an effective wax remover which has no detrimental effects on the panel surface (see 3.6), and are sponged and rinsed freely. This procedure may be repeated once, but the panel is rejected for use if it is not cleaned this time. Water will not run off the panel in droplets as it did from the finished test panels above, but the surface should be relatively uniform in color with no trace of dyed wax on the surface. After thorough rinsing the panel is allowed to drain and dry in a vertical position in a drain rack. A coat of plain 16 percent test wax (see 4.4.5.2.2) is applied to the panel. The panel is immediately placed on edge in a drain rack. This allows the excess wax to flow off and a thin uniform layer of wax is left on the panel. The panels are dried in an oven at 50° ± 2°C. for approximately 48 hours. After they have cooled to room temperature, these panels are treated with 5 percent potassium oleate soap solution, washed and dried as described above for new panels.

The amber reflectance of these panels are then determined with a Hunter Multi-Purpose Reflectometer and are recorded as R_1. The reflectometer should be standardized at the amber reflectance closest to the amber reflectance of the linoleum panels. Four duplicate panels should be prepared for each test sample for evaluation. These panels should be read not more than one day before the application of the dyed wax. A slight change (usually an increase) may be observed on a prolonged standing, but this is rendered almost negligible by the heat setting at 50°C. in the above procedures. A light pencil guide line is useful for alignment of the dyed wax film with the reflectometer opening. With an application along this guide line, the same reading positions may be resumed for each of the various reflectance readings on each panel.

4.4.5.2.2 The test wax shall be formulated in the following manner:

Basic Charge for 40 Gallons of Resin Dispersion

Materials	*Specification*	*Quantities*
Ammonia	– – – – –	4 pounds
Morpholine	– – – – –	4 pounds
Triethanolamine	– – – – –	4 pounds
Loba "B" or Loba "C" resin	Softening Point—73° = 90°C, Melting Point 114°-124°C.	35 pounds
Water	– – – – –	Sufficient to make 40 gallons

Procedure for processing

Starting with 20 gallons of tap water, under agitation, add 4 lbs. of ammonia, 4 lbs. of morpholine and 4 lbs. of triethanolamine. This mixture should be agitated for not less than 15 minutes. Add 35 lbs. pulverized loba "C" resin. This mixture should be agitated for not less than 30 minutes, then heated to a temperature of 131 degrees F. and kept at this temperature for 10 minutes. Add 20 gallons of cold tap water and agitate for 5 minutes.

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Basic Charge for 60 Gallons of Water Emulsion Wax

Materials	*Specification*	*Quantities*
Carnauba wax	Commercial, "North Country" Grade No. 3, refined, Federal Specification JJJ–W–141. (Softening point 83°C.)	36 pounds
Microcrystalline wax	190°-195°F. (c.p.)	4 pounds
Oleic acid	Titer: 10°-12°C. F.F.A. as oleic; 95-98 Acid No. 189-195 Saponification value: 195-200 Iodine value: 90-93	8 pounds
Morpholine		5 pounds
Sodium alkyl sulfate (Tergitol #4)		1.5 pounds
Sodium salt of alkyl aryl polyether sulfate (Triton W-30) [1]		9.6 pounds
Resin dispersion*		8 gallons
Colloidal silica, 30 percent		72 pounds
Water		Sufficient to make 60 gallons

[1] Obtainable from Rohm and Haas Co., Phila., Pa.

**Note.*—8 gallons of Resin Dispersion to be used in making 60 gallon batch of water emulsion wax.

Procedure for processing

Melt 4 lbs. of microcrystalline paraffin wax, adding 36 lbs. of carnauba wax and 8 lbs. of oleic acid, heating the solution to approximately 210°F. Then add 4 lbs. morpholine and let mixture be agitated for not less than 1 minute with a wooden hand paddle. Add 1½ lbs. sodium alkyl sulfate (Tergitol #4) and agitate for not less than 3 minutes and bring temperature to 212°F. (this temperature is important). Very rapidly (less than 30 seconds) add 30 gallons of water having a temperature of 212°F. (this temperature is also very important). When mixture drops to between 208° and 212°F., it should be agitated for not less than 5 minutes, and then allowed to cool to 110°F. while still under agitation. Add 9.6 lbs. of sodium salt of alkyl aryl polyether sulfate (Triton W-30) and agitate for 3 minutes. Add 8 gallons resin dispersion and agitate for 5 minutes. Finally add 72 lbs. of 30 percent colloidal silica and agitate for 10 minutes.

Before the material is placed in containers, it should be filtered through either 100-mesh screen or cheesecloth.

The special formulation of wax for this test is a standard grade of liquid wax which is defined chemically and has a narrow range of variations in physical and chemical properties. Waxes with no set chemical and physical formulations have a wide range of chemical and physical properties which vary with different batches and different manufacturers. Only experienced wax formulators should prepare this liquid wax for these tests (even though the expense is greater for a special test product), since satisfactory adherence to the described formulation is an art derived from experience in the wax field.

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4.4.5.2.3 *Dyeing of wax.*—To 100 ml. of the wax as prepared in 4.4.5.2.2 add 0.200 grams of indulin bluish dye, C.I. 861. Stir until completely dissolved. Filter through cheesecloth. A new dyed-wax solution must be prepared every 30 days since the dye is not completely stable in wax solutions.

4.4.5.2.4 *Test applicator.*—The test applicator shall be a piece of rigid plastic sheet, 3 inches square and 3/16 inch thick. One end-side of the applicator shall be rounded off on both edges of the side, 1/16 inch down from the edges. Exactly 1/16 inch in from the ends of the rounded side, the side shall be recessed to provide a flat surface 1/16 inch in width, and 0.008 inch clearance between the recessed surface and a test panel.

4.4.5.2.5 *Application of dyed test wax.*—The physical properties of liquid wax prevent any single draw-down application with standard film thickness blades. A draw-down applicator must be moved slowly to give uniform spreading and under these conditions there is a certain amount of flowback under the blade clearance. It takes about 30 seconds for a wet wax film to wet the surface of the finish surface of the linoleum.

During this time a puddling effect is displayed by a tendency for drop formation from the cohesive properties of the dyed wax. This puddling effect is sufficiently great when panels are not free of surface film to give large bare areas in the dyed wax film almost instantaneously. Exact delivery volume is useless, however, additional dyed wax requires more leveling strokes and more draw-off at the edge of the panel. Some of the dyed wax also adheres to the applicator and the amount is dependent upon the spread of the drops and other physical environmental conditions. More satisfactory spreading is obtained by spacing the separate drops in a rectangular pattern slightly smaller than the 3-inch x 6-inch desired film. The exact number of drops (12-14) depend somewhat on environmental conditions of the day.

Place the test panel on a level table surface, lay the beveled edge of a ruler along the panel guide line, and add approximately twelve drops from an eyedropper in a rectangular pattern such as three lines of four drops of the dyed test wax as prepared in 4.4.5.2.3. Touch the center of the rounded side of the test applicator to the first row of three drops of dyed wax which is perpendicular to the ruler, and allow the drops to run together to fill the entire clearance of the recessed surface. With the ruler as a guiding edge, hold the applicator in a vertical position and with several back and forward strokes, slowly spread the wax over the 3 inch x 6 inch of surface adjacent to the edge of the panel. The applicator is wiped dry with a cloth and the procedure is repeated on a second panel. In the meantime, the film of the first panel has lost its uniform appearance within the first fifteen seconds. The ruler is again placed along the guide line of this first panel sometime between 40 and 80 seconds, depending upon the drying conditions, and the applicator is placed in position above the heavier at the first draw-off position of the initial spreading and is slowly moved back and forth across the panel until a uniform film is obtained, and it is again drawn off at the end of one of these strokes when the film is uniform. The thicker film at the draw off edge is removed immediately with a cloth. This prevents its flow into the central portion of the panel where the measurements are made. The second panel is now completed in like manner with a dried applicator. The applicator should be washed and wiped dry after the completion of each second panel. This prevents solidification and change in the special rounded applicator surface. Allow the resultant film to air dry in the same horizontal position until the film has a completely dry appearance.

Allow the resultant film to air dry at room temperature in racks as described in 4.4.5.-2.1. The amber reflectance of the dyed wax film shall be determined on the 8th day and recorded as R_2. The daily count shall begin on the day of the film application. The re-

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flectometer should be standardized at the amber reflectance closest to the amber reflectance of the dyed wax film. The efficiency procedure shall be run on the eighth day.

4.4.5.3 *Preparation of samples of wax removers.*

Type I.—To prepare a solution of the compound, add 100 ml. of the sample from 4.4.1 to a graduate containing approximately 400 ml. of distilled water, mix and then dilute to 900 ml. with distilled water. The solution shall be used at 70°-75°F. for these tests.

Types II and III.—To prepare a solution of the compound, add 30 ml. of the sample from 4.4.1 to a graduate containing approximately 400 ml. of distilled water, mix and then dilute to 990 ml. with distilled water. The solution shall be used at 70°-75°F. for these tests.

4.4.5.4 *Procedure.* — Place the dyed wax panel in the washability tray in such a position so that the length of the dyed wax film is parallel to the motion of the sponge. The sponge should then pass over practically all of the film and insure uniform removal in the region for the three reflectance readings. (There should be no directional difference in the dyed wax film since it is a free flowing film). Place metal or other suitable panels on both sides of the test panel to hold it in place in the middle of the washability tray and to provide a level surface for the sponge to travel over. Insert sponge into sponge box and add sufficient solution with a temperature of 70°-75°F. as prepared in 4.4.5.3 to immerse the panel surface under at least ⅛ inch of diluted wax remover. The washing procedure should be completed with the solution in the range of 70°-75°F. (A specific volume which would be dependent upon specification leveling panels, is not necessary and excess coverage depth above ⅛ inch does not change the effective removal efficiency.) The sponge absorbs removal solution as it is added to the tray. Allow the panel to soak for 4½ minutes. Start the mechanism and allow to continue for 40 strokes (20 complete oscillations as recorded on the counter). Remove the panel and immediately rinse with clear, cold tap water until it flows off freely in droplets. Allow the panel to completely dry at room temperature and measure the resultant amber reflectance of the panel and record as R_3. This reading should be made preferably on the eighth day, but not later than the ninth day. Calculate the wax removal efficiency by the following methods:

$$\text{Wax removal efficiency percent} = 100 \times \frac{(R_3 - R_2)}{(R_1 - R_2)}$$

Where:

R_1 is amber reflectance of new panel.

R_2 is amber reflectance of waxed panel.

R_3 is amber reflectance of stripped panel.

Make four determinations of the removal efficiency and average the results obtained on at least three of these panels. Panels which cannot be stripped of dyed wax in the processing for reuse as described in 4.4.5.2.1, shall not be used in the "efficiency average" above.

4.4.6 *Deleterious action on linoleum, asphalt, and rubber tile flooring.*

4.4.6.1 *Test panels.*—These panels shall be as specified below:

Linoleum panels—2 inch x 4 inch x ⅛ inch, conforming to Federal Specification LLL–L–367.

Asphalt panels—2 inch x 4 inch x ⅛ inch, conforming to Federal Specification SS–T–306.

Rubber tile panels—2 inch x 4 inch x ⅛ inch, conforming to Federal Specification ZZ–T–301.

4.4.6.2 *Procedure.*—Mix 50 ml. of type I with 400 ml. of distilled water or 20 ml. of types II and III with 640 ml. of distilled water. Pour 150 ml. of the solution into each of three 250-ml. beakers. Immerse one panel of each type in the beakers which contain the wax remover solutions. Allow the panels to

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soak for 16 hours at 70°-80°F. (room temperature). At the end of this time, remove the panels, rinse thoroughly with distilled water and dry at room temperature for at least one hour.

Examine the panels which have assumed a completely dry appearance for objectionable deleterious actions. There shall be no signs of softening, blistering, cracking, pitting, or any difficulty soluble deposits on the surface. The solution shall not bleach, stain, oxidize or change the color of the panels in any form to any appreciable degree. (Notes: Most satisfactory removers show a slight lightening of color between the exposed and immersed surfaces. This should not constitute a failure for the test.)

4.4.7 *Volatile ammonia (type III only).*—Heat 150 ml. of the product diluted 1:32 to boiling. Suspend a piece of moist red litmus paper in the vapors approximately one inch over the surface of the boiling solution. Any change in the color of the red litmus to blue shall be cause for rejection.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.—Packaging shall be level A, B, or C as specified.

5.1.1 *Levels A and B.*

5.1.1.1 *Unit containers.* — Wax remover shall be furnished in 1-gallon containers which shall conform to type V, class 2, class 4, or class 5 of Federal Specification PPP–C–96, with exterior coating, plan B for level A and plan A for level B.

5.1.1.2 *Bulk containers.* — Wax remover shall be furnished in 5-gallon or 55-gallon quantities, as specified in the contract or order (see 6.2). Five-gallon containers shall conform to class 3, type I of Federal Specification PPP–P–704 with container equipped with spout or nozzle closure, with outer seal. Color of exterior coating shall be olive drab. Fifty-five-gallon drums shall conform to either type II or type IV of Federal Specification PPP–D–729.

5.1.1.3 Containers manufactured from metals which are susceptible to corrosive reaction by any ingredient in the product shall be coated on the inside with a coating which shall neither affect nor be affected by the product packed. Contractor shall certify to the procuring agency that the coating used complies with this requirement.

5.1.2 *Level C.*—The product shall be packaged in accordance with commercial practice.

5.2 Packing.—Packing shall be level A, B, or C as specified.

5.2.1 *Level A.*

5.2.1.1 *Unit containers.* — Wax remover, packaged in 1-gallon containers, shall be packed in accordance with the appendix to Federal Specification PPP–C–96 for overseas shipment.

5.2.1.2 *Bulk containers.* — Five-gallon drums or pails, or 55-gallon drums shall require no overpacking.

5.2.2 *Level B.*

5.2.2.1 *Unit containers.* — Wax remover, packaged in 1-gallon containers, shall be packed for shipment in accordance with the appendix to Federal Specification PPP–C–96 for domestic shipment.

5.2.2.2 *Bulk containers.* — Five-gallon drums or pails, or 55-gallon drums require no overpacking.

5.2.3 *Level C.*—Commercial unit and bulk containers shall be packed so as to be acceptable by common or other carriers for safe

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transportation to point of destination specified in shipping instruction at the lowest transportation rate.

5.3 Marking.

5.3.1 *Civil agencies.*—In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*— In addition to any special marking required by the contract or order, marking of shipments shall be in accordance with Military Standard MIL–STD–129.

5.3.3 *Directions for use.*—The contractor shall mark each unit and shipping container with a complete set of clear and concise instructions on the methods recommended in obtaining satisfactory results with the product in floor maintenance operations. A statement shall be included regarding care to be exercised for prevention of possible skin irritation when using the item.

6. NOTES

6.1 Intended use.—The compound covered by this specification is intended for use in removing films of water emulsion wax from floors.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following: (See 1.2, 5.1, 5.2, 5.3, 6.2.1 and 6.4).

6.2.1 In procuring the material interchangeable as to type, the bids should be computed as follows for uniformity in arriving at low bid:

For type I, bid price per gallon ÷ 9 = basis of award price.

For type II, bid price per gallon ÷ 33 = basis of award price.

6.3 Transportation description.—Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Cleaning compound, not otherwise indexed by name, liquid.

Carload minimum weight 36,000 pounds.

Motor:

Cleaning compound, not otherwise indexed, liquid.

Truckload minimum weight (W) 36.6 pounds, (W) subject to Rule 34, National Motor Freight Classification.

6.4 The following statement should be included in the purchase contract: "If latent defects should be discovered for a period of 9 months after material has been accepted, the Government reserves the right to reject the delivery, require replacement, or cancel the contract and surcharge the contractor for any expense incurred by the cancellation of the contract and in securing satisfactory material."

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M Q
Navy—Sh
Air Force—MOA.

Copies of this specification may be purchased for 7 cents each.

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